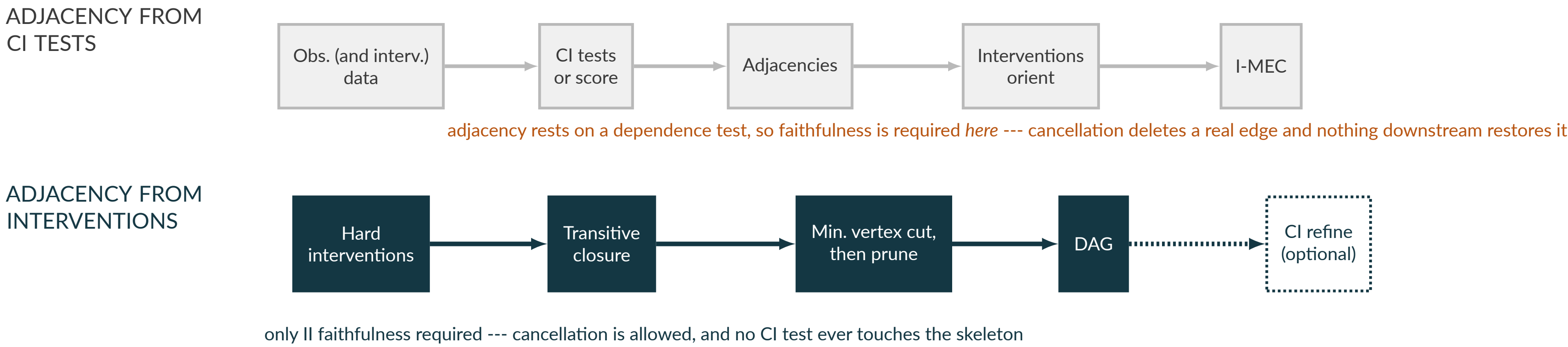




Flipping the Causal Discovery Workflow



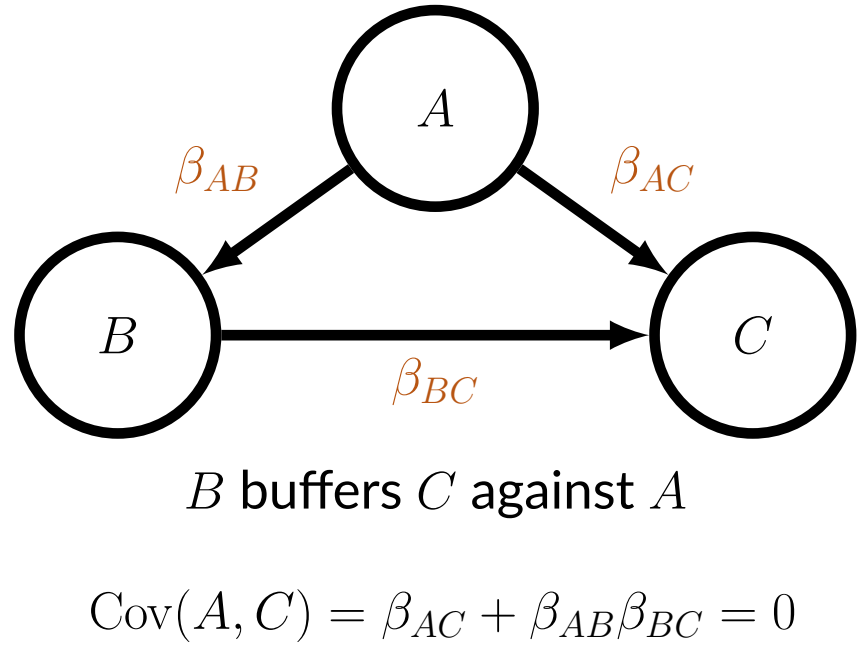
To be clear about what is and is not new: plenty of methods already fold interventional data in *jointly* rather than sequentially --- GIES, IGSP and variants, JCI, DCDI. The distinction drawn here is not *when* interventions enter, but what they are *for*. Those methods still source **adjacency** from a dependence test or a score requiring (λ -strong) faithfulness, so path cancellation removes an edge that no later step recovers. GSP-style permutation search is the closest precedent, and remains CI-based.

Main Result

Hard interventions reveal whether an edge *exists*, not merely which way it points. Under **intervention-immediacy (II) faithfulness** --- which *tolerates* path cancellation --- the DAG is nonparametrically identifiable from interventions alone.

Cancellation, Change Sets, and Immediacy

Cancellation Is a Design Feature



Constraint-based discovery assumes **faithfulness**: causally linked variables are dependent. Biological systems break this on purpose, since pathways are **buffered** by compensating routes [Hartman et al., 2001].

A CI test reads that zero as "no edge" and deletes $A \rightarrow C$; nothing downstream puts it back. Cancellation also defeats *interventional* faithfulness [Chevalley et al., 2025].

``But isn't exact cancellation measure zero?"

Under *exact* statistics, yes --- faithfulness holds with full Lebesgue measure [Meek, 1995]. Finite samples change the question. There you need λ -strong faithfulness, so what matters is not whether the parameters land exactly on a cancellation but how *close* they sit to one. That is where the geometry bites →

What an Intervention Reveals

Let $I(\mathbf{A})$ be the simultaneous **hard** intervention on $\mathbf{A}$ --- a knockout, which severs each target from its parents.

$$\text{CHG}(I(\mathbf{A})) := \{V : P_{I(\mathbf{A})}(v) \neq P(v)\}$$

The **conditional** change set asks what a second knockout moves once a first is held fixed --- the logic of sequential gene knockouts, where an earlier knockout $I(\mathbf{B})$ strips away upstream regulation so that the effect of $I(\mathbf{A})$ can be attributed to the paths that remain:

$$\text{CHG}(I(\mathbf{A}) \mid I(\mathbf{B})) := \{V : P_{I(\mathbf{A} \cup \mathbf{B})}(v) \neq P_{I(\mathbf{B})}(v)\}$$

Two Bounds on a Change Set

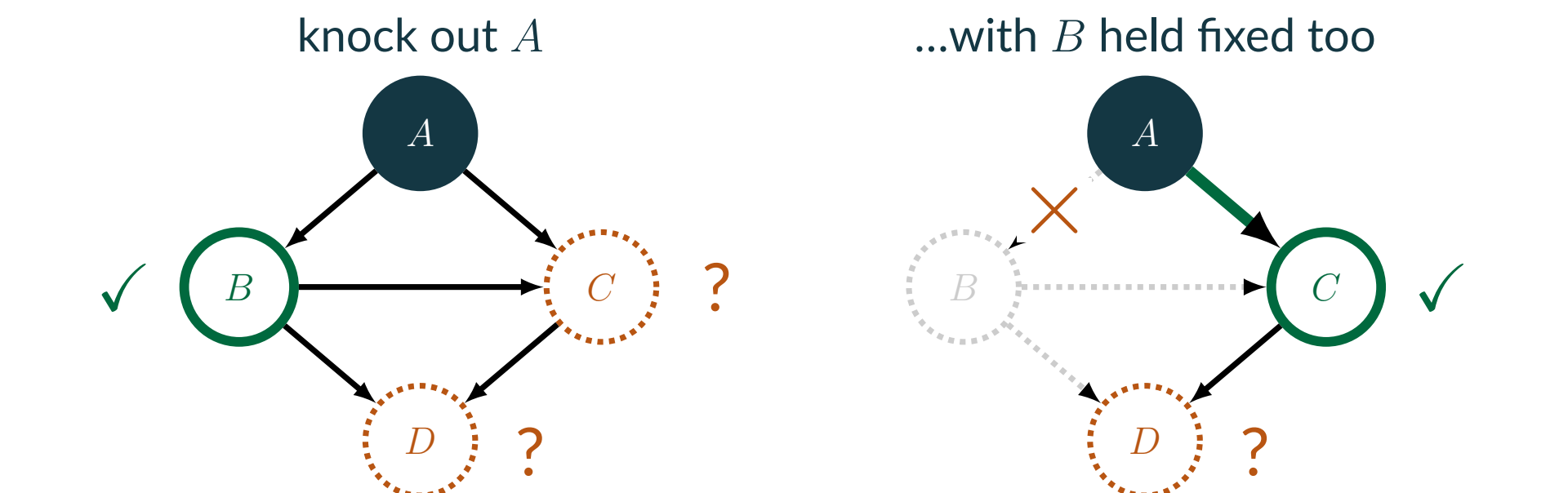
$$\underbrace{\text{MIC}_{\mathcal{G}}(A)}_{\text{II faithfulness}} \subseteq \text{CHG}(I(\{A\})) \subseteq \underbrace{\{A\} \cup \text{DE}_{\mathcal{G}}(A)}_{\text{intervention locality}}$$

Locality caps a change set from above. It is not an assumption --- the SCM implies it, exactly as it implies the causal Markov condition.

Immediacy pins it from below, and this one must be assumed. It is the whole content of the paper's relaxation.

Immediacy Beats Cancellation

Most immediate child (MIC). B is a MIC of A when $A \rightarrow B$ is the *only* directed path from A to B . Every non-sink has one.



✓ **guaranteed to change** --- reached by one path only. ? **may change** --- competing paths can cancel, so no guarantee either way.

Conditional MIC. A second intervention breaks the arrows into its own targets. With B held fixed, the route $A \rightarrow B \rightarrow C$ is gone, so $A \rightarrow C$ stands alone and C joins the guaranteed set. **Immediacy is manufactured on demand.**

II Faithfulness

For all disjoint $\mathbf{A}, \mathbf{B}$, every conditional MIC of every $A \in \mathbf{A}$ given $I(\mathbf{B})$ lies in $\text{CHG}(I(\mathbf{A}) \mid I(\mathbf{B}))$.

A Milder Assumption, and Two Algorithms That Use It

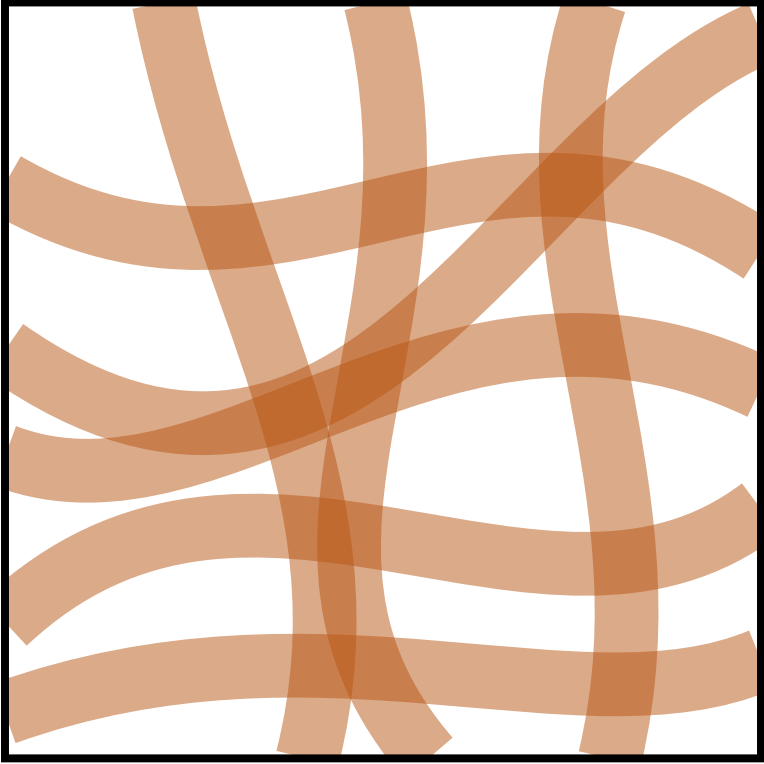
How Mild Is II Faithfulness?

In a linear SEM with unit-variance noise, the strong version is purely **local**: it constrains single edges, never sums or products along competing paths.

Lemma

A model is λ -strong mean II faithful iff $|\beta_{ij}| \geq \lambda$ for every edge. So the II-violating parameterizations are a *strict subset* of the CI-violating ones.

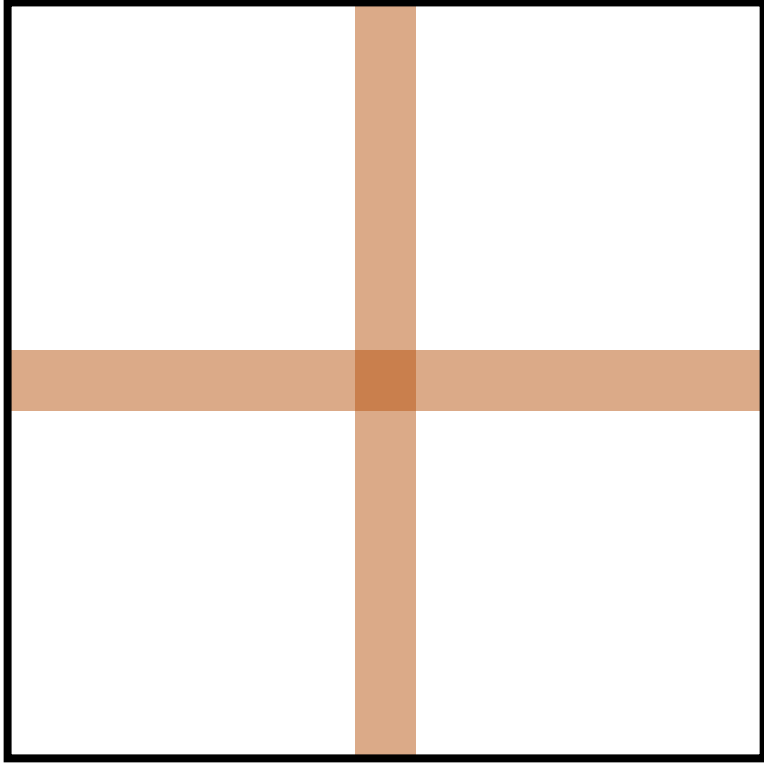
λ -strong CI faithfulness



violations fill the cube

$$\omega(\lambda^{\text{poly}(n)} 2^{|E|})$$

λ -strong mean II faithfulness

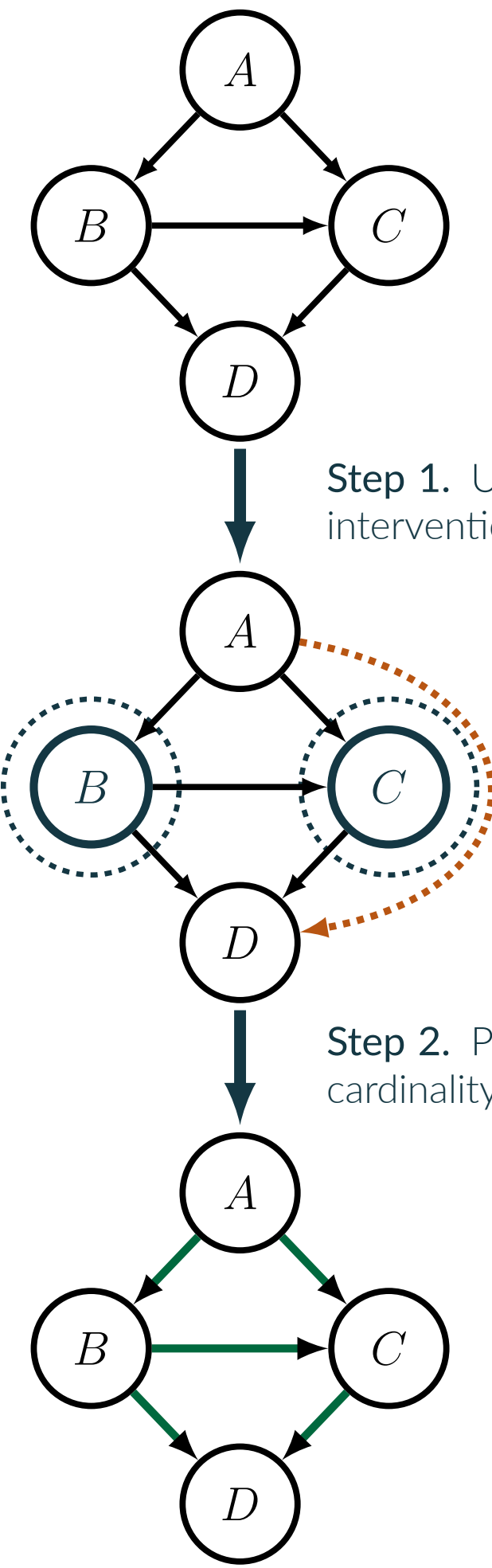


violations are thin slabs

$$\mathcal{O}(\lambda |E|)$$

The violation volume grows only **linearly** in λ , against the space-filling behaviour Uhler et al. (2013) established for CI faithfulness.

What The Interventions Recover



true DAG $\mathcal{G}$

What we want. Unknown at the start --- and unreachable by CI tests when paths cancel.

Step 1. Union the change sets of the single-node interventions $I(V)$.

TC($\mathcal{G}$)

Exactly the transitive closure: *no true edge is ever missing*, but the ancestral shortcut $A \rightarrow D$ is along for the ride.

Only $\mathbf{Z} = \{B, C\}$ separates A from D , so $\kappa_{\max} = 2$. This graph is therefore *also* the k -robust closure for every $k \leq 2$.

Step 2. Prune $A \rightarrow D$ by conditioning on $I(\mathbf{Z})$ --- cardinality $3 > \kappa_{\max}$.

$\mathcal{G}$ recovered

Exact, with $\mathcal{O}(n^2)$ interventions, once the budget clears the connectivity: $k > \kappa_{\max}$.

Below it, k -RIC returns the k -robust closure above instead.

Algorithms

Walk the closure in topological order and test each candidate edge under a conditioning intervention chosen to kill every competing path.

UIC --- unrestricted cardinality

```
1: for each  $V_j$  in a topological order of  $\mathcal{G}'$  do
2:   for each  $V_i \in \text{PA}_{\mathcal{G}'}(V_j)$  do
3:      $\mathbf{B} \leftarrow \text{PA}_{\mathcal{G}'}(V_j) \setminus \{V_i\}$ 
4:     if  $V_j \notin \text{CHG}(I(\{V_i\}) \mid I(\mathbf{B}))$  then
5:       remove  $V_i \rightarrow V_j$ 
6:     end if
7:   end for
8: end for
```

Theorem. Recovers $\mathcal{G}$ with $\mathcal{O}(n^2)$ interventions --- but of cardinality up to $n - 1$.

Nobody knocks out $n - 1$ genes at once. So cap cardinality at k and condition on a **minimum vertex cut** instead of all parents --- searched for on the fly, making the method **adaptive**.

k -RIC --- cardinality budget k

```
1:  $\mathcal{G}'' \leftarrow \mathcal{G}'$  with  $V_i \rightarrow V_j$  removed
2:  $\mathbf{Z} \leftarrow$  min vertex cut separating  $V_i, V_j$  in  $\mathcal{G}''$ 
3: if  $|\mathbf{Z}| < k$  and  $V_j \notin \text{CHG}(I(\{V_i\}) \mid I(\mathbf{Z}))$  then
4:   remove  $V_i \rightarrow V_j$ 
5: end if
```

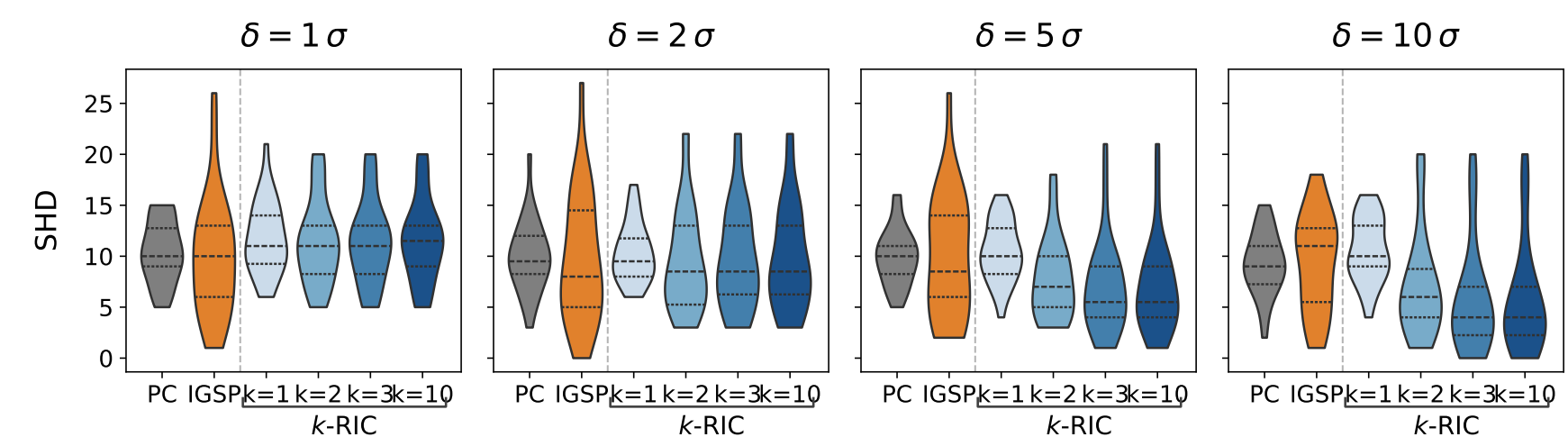
Theorem. If $k > \kappa_{\max}$, the largest local vertex connectivity over non-adjacent pairs, k -RIC recovers $\mathcal{G}$ exactly with $\mathcal{O}(n^2)$ interventions.

Below that threshold, structure is identified up to the **k -robust transitive closure** above. A permutation is its weakest, 0-robust case, placing permutation-based discovery [Wang et al., 2017] at one end of this spectrum.

Intervention-Only Discovery Beats CI-Based Discovery --- and Survives Cancellation

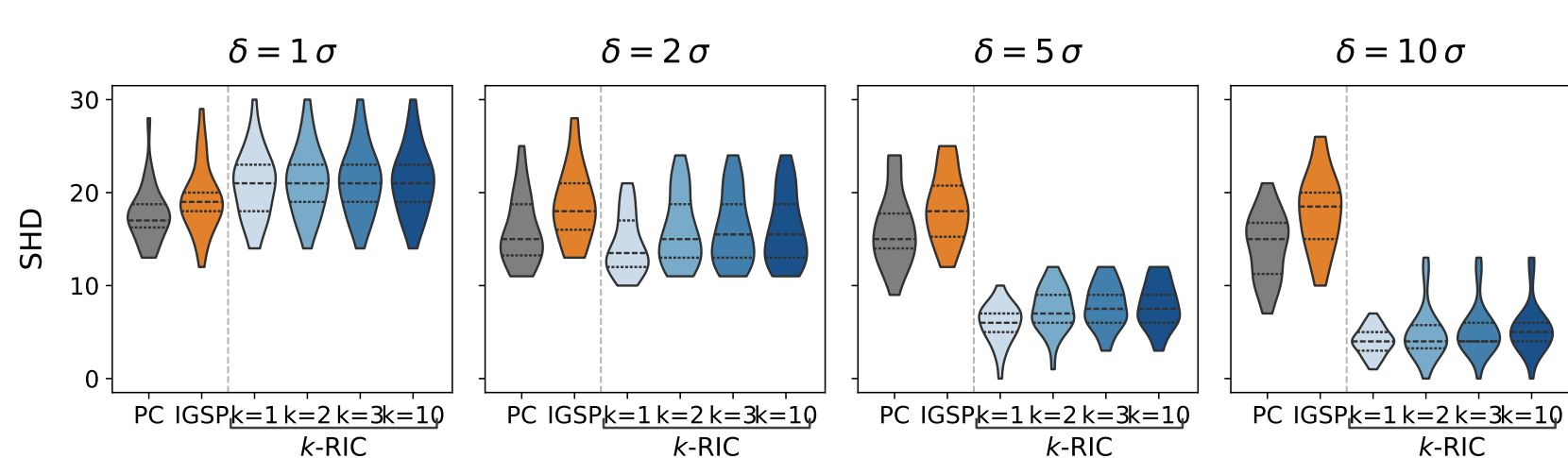
Setup. 30 random DAGs, 10 nodes, 25 expected edges, standard Gaussian noise; 1000 observational samples and 1000 per intervention. A knockout resets its target to a Gaussian shifted by δ noise SDs, detected by a two-sample t -test at $\alpha = 0.05$, Bonferroni-corrected. Baselines **PC→Interv** [Spirtes et al., 2000] and **IGSP** [Wang et al., 2017] get the same single-node data. Lower SHD is better; $k = 10$ is unrestricted and coincides with UIC.

Baselines (PC→Interv, IGSP) vs k -RIC across cardinality $k=1, 2, 3, 10$ (Nodes=10, Edges=25, weights in (-1, 1))



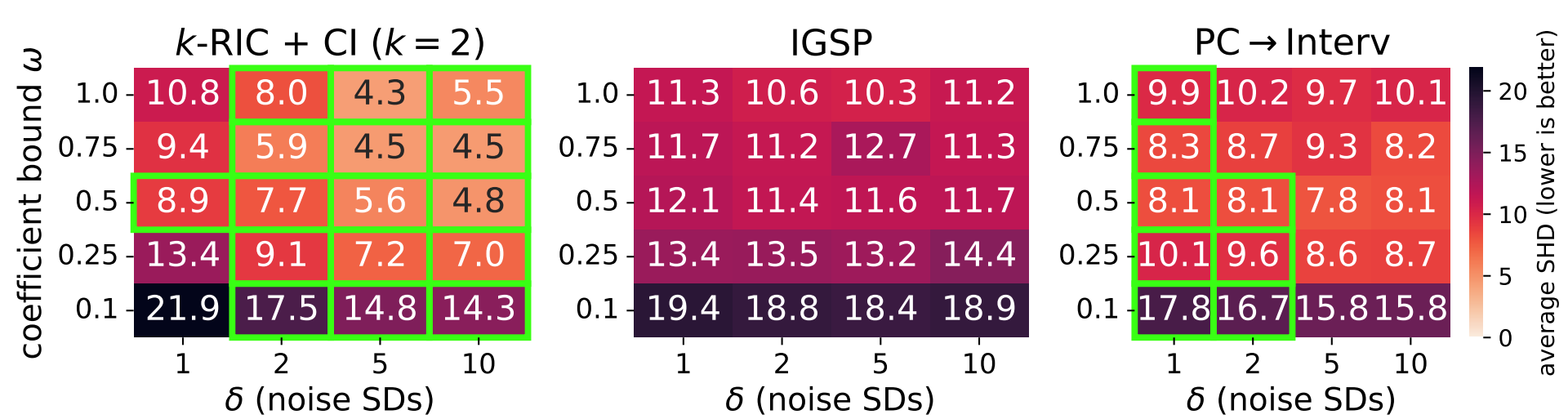
Standard regime ($\beta \in [-1, 1]$). Cardinality buys accuracy: $k = 1$ carries the closure's spurious ancestral edges and multi-node interventions prune them. k -RIC passes both baselines by $k = 2$ for $\delta \geq 5\sigma$, reaching SHD ≈ 4 at $\delta = 10\sigma$.

Baselines (PC→Interv, IGSP) vs k -RIC across cardinality $k=1, 2, 3, 10$ (Nodes=10, Edges=25, weights in (-0.1, 0.1))



Cancellation regime ($\beta \in [-0.1, 0.1]$) --- the stress test. Sign-varied small coefficients cancel partial correlations, so both baselines stay near-uninformative at every δ (SHD ≈ 15 --19), while k -RIC reaches SHD ≈ 4 --5 at $\delta = 10\sigma$.

Average SHD by method (green box = best, or tied within 'se')



Across the plane. Mean SHD over 20 DAGs per cell, sweeping the coefficient bound ω against strength δ ; green marks best or within one SE of best. The intervention-first pipeline wins nearly everywhere, ceding only the weakest-signal corner, where the induced shift ($\propto \omega\delta$) cannot resolve structure.

What To Take Away

- Hard interventions identify **edge presence**, not only orientation --- the CI-first hierarchy is not forced on us.
- The assumption that buys this is **edge-local**, and its violation set shrinks *linearly* in λ .
- Identifiability turns on one graph quantity, $\kappa_{\max}$, against one experimental quantity, the budget k .
- A method using **no CI tests at all** matches or beats methods that use them. So build assays for **multi-node** perturbation.

References

- Chevalley, M., Schwab, P. & Mehrjou, A. Deriving causal order from single-variable interventions: guarantees & algorithm. *ICLR*, 2025.
- Hartman, J. L., Garvik, B. & Hartwell, L. Principles for the buffering of genetic variation. *Science* 291(5506):1001--1004, 2001.
- Meek, C. Strong completeness and faithfulness in Bayesian networks. *UAI*, 1995.
- Spirtes, P., Glymour, C. & Scheines, R. *Causation, Prediction, and Search*. MIT Press, 2000.
- Uhler, C., Raskutti, G., Bühlmann, P. & Yu, B. Geometry of the faithfulness assumption in causal inference. *Ann. Statist.*, 436--463, 2013.
- Wang, Y., Solus, L., Yang, K. & Uhler, C. Permutation-based causal inference algorithms with interventions. *NeurIPS* 30, 2017.

Limitations

Causal sufficiency is assumed; extending II faithfulness to ancestral graphs with latents is open. Hard interventions must be available --- natural for knockouts, not for much of environmental or social science. Our tests detect mean shifts, so they certify only *mean* II faithfulness; a nonparametric test recovers the distributional notion. Soft interventions never sever a node from its parents, so they need a stronger assumption.

Thanks

Published as: Mazaheri, Zhang & Uhler. Relaxing Faithfulness with Intervention-Only Causal Discovery. *Proc. 42nd Conf. on Uncertainty in Artificial Intelligence*, PMLR 337:4441--4456, 2026.

B.M. is partially supported by the Advanced Research Concepts (ARC) COMPASS program, sponsored by DARPA under agreement HR001-25-3-0212. J.Z. was partially supported by the Eric and Wendy Schmidt Center. C.U. was partially supported by NCCIH/NIH (1DP2AT012345), ONR (N00014-24-1-2687), DOE-ASCR (DE-SC0023187), and the Eric and Wendy Schmidt Center at the Broad Institute.